EASY · PROBLEM-SOLVING AND DATA ANALYSIS

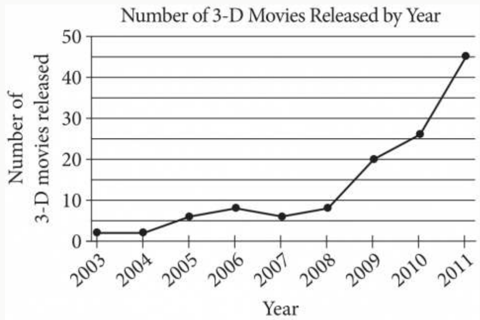
Two-variable data: Models and scatterplots

03

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS



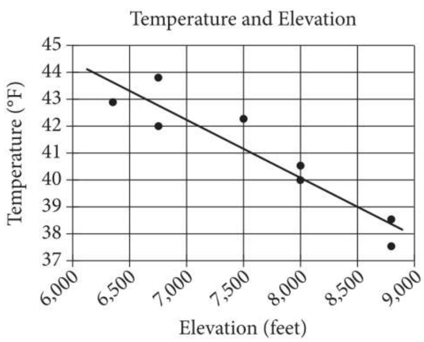
According to the line graph above, between which two consecutive years was there the greatest change in the number of 3-D movies released?

A. 2003–2004

B. 2008–2009

C. 2009–2010

D. 2010–2011



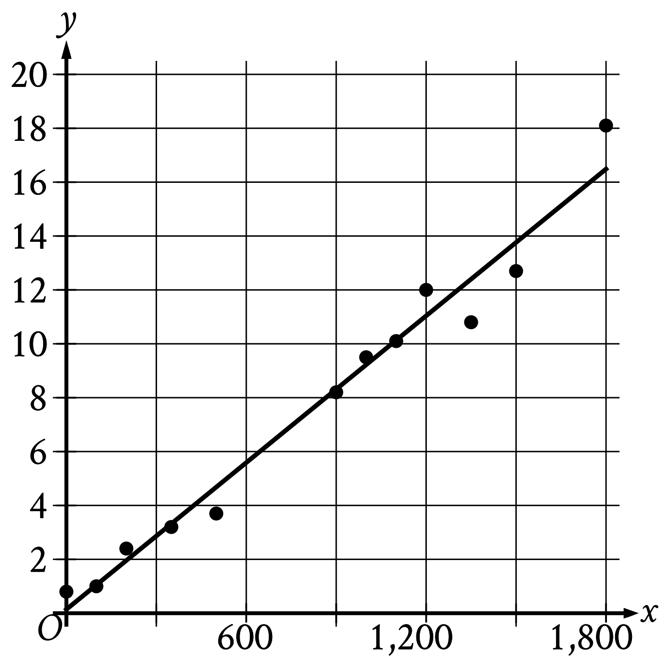
The scatterplot above shows the high temperature on a certain day and the elevation of 8 different locations in the Lake Tahoe Basin. A line of best fit for the data is also shown. What temperature is predicted by the line of best fit for a location in the Lake Tahoe Basin with an elevation of 8,500 feet?

A. 37°F

B. 39°F

C. 41°F

D. 43°F



- The scatterplot has 12 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (600 comma 5.6)
    - (1,200 comma 11.0)
    - (1,500 comma 13.8)
    - (1,800 comma 16.5)

Twelve data points are shown in the scatterplot. A line of best fit for the data is also shown. At $x = 1{,}200$, which of the following is closest to the y-value predicted by the line of best fit?

A. 16

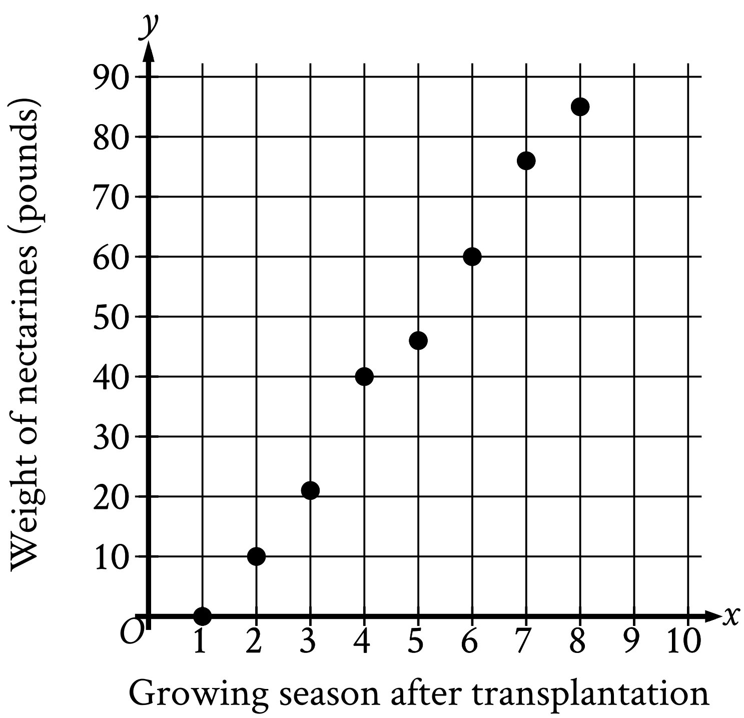
B. 14

C. 11

D. 6

Q4 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

An orchard owner recorded the weight, in pounds, of all nectarines that grew on a dwarf nectarine tree during each growing season after the tree's transplantation. The scatterplot shows this weight, in pounds, for each growing season after the tree's transplantation.



- The scatterplot has 8 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 0)
  - (2 comma 10)
  - (3 comma 21)
  - (4 comma 40)
  - (5 comma 46)
  - (6 comma 60)
  - (7 comma 76)
  - (8 comma 85)

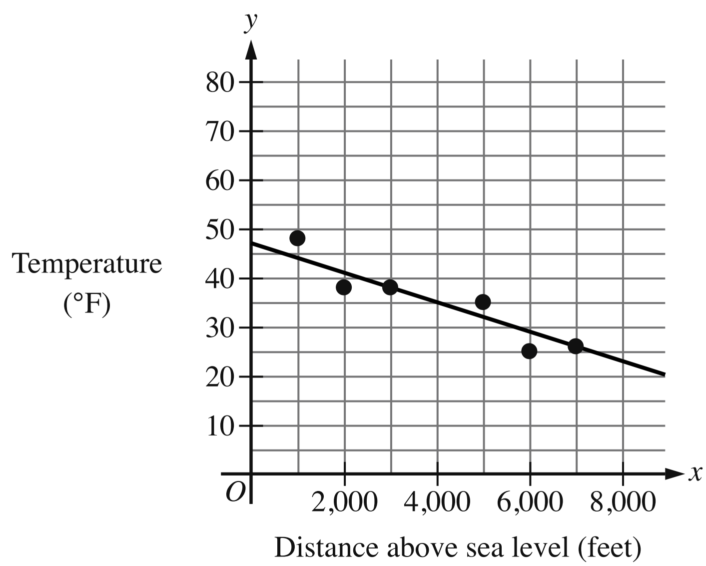
What was the weight, to the nearest pound, of all nectarines that grew on the tree during the 4th growing season after the tree's transplantation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the temperature, in degrees Fahrenheit °F, and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
    - The line of best fit slants down from left to right.
    - The line of best fit passes through the following coordinates:
        - (0 comma 47)
        - (4,000 comma 35)
        - (7,000 comma 26)

At a distance of 4,000 feet above sea level, what is the temperature, in °F, predicted by the line of best fit?

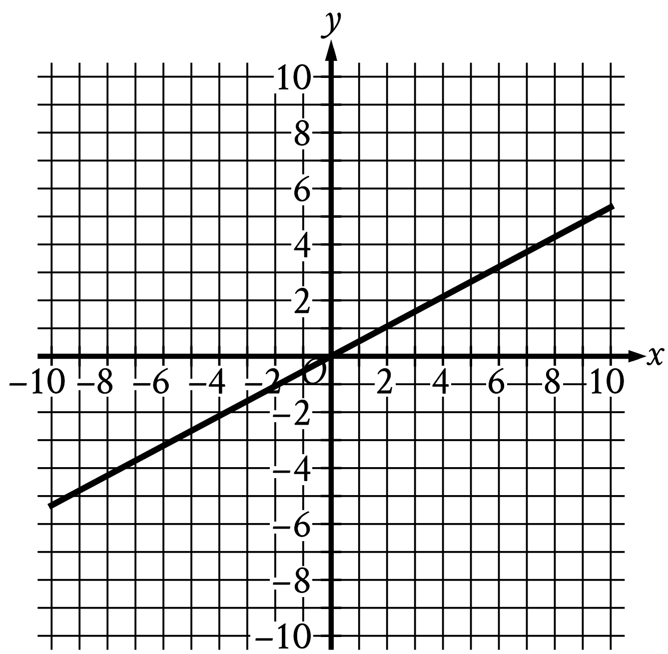
A. 47

B. 35

C. 25

D. 0

The graph of function f is shown, where $y = fx$.



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 6 comma negative StartFraction 16 Over 5 EndFraction)
  - (0 comma 0)
  - (6 comma StartFraction 16 Over 5 EndFraction)

Which of the following describes function f?

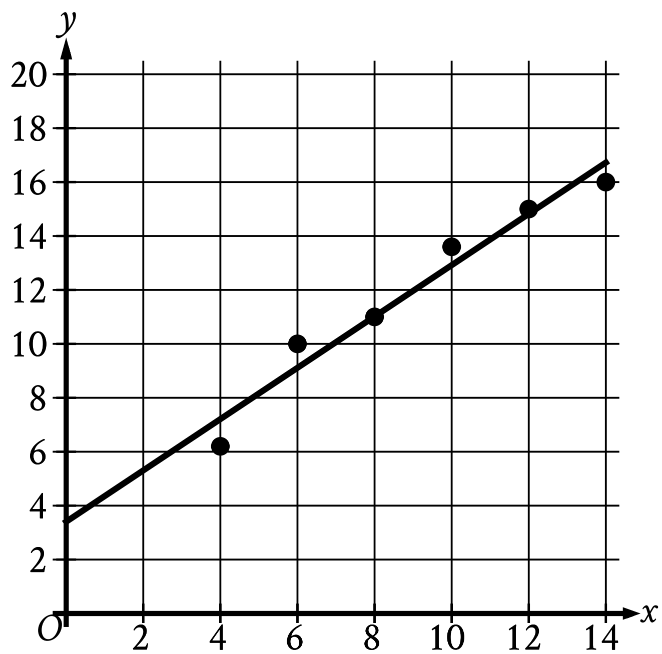
A. Increasing linear

B. Decreasing linear

C. Increasing exponential

D. Decreasing exponential

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 3.4)
    - (8 comma 11)
    - (12 comma 14.8)

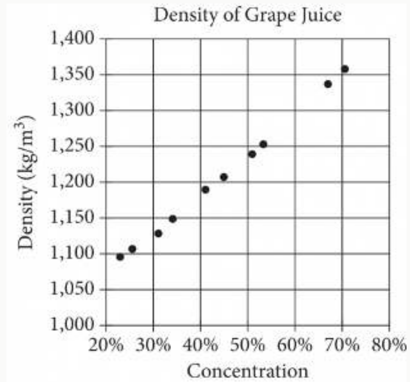
Which of the following equations best represents the line of best fit shown?

A. $y = x + 3.4$

B. $y = x - 3.4$

C. $y = -x + 3.4$

D. $y = -x - 3.4$



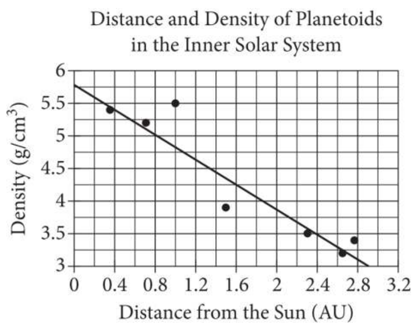
The densities of different concentrations of grape juice are shown in the scatterplot above. According to the trend shown by the data, which of the following is closest to the predicted density, in kilograms per cubic meter (kg/m^3), for grape juice with a concentration of 60%?

A. 1,200

B. 1,250

C. 1,300

D. 1,350



The scatterplot above shows the densities of 7 planetoids, in grams per cubic centimeter, with respect to their average distances from the Sun in astronomical units (AU). The line of best fit is also shown. An astronomer has discovered a new planetoid about 1.2 AU from the Sun. According to the line of best fit, which of the following best approximates the density of the planetoid, in grams per cubic centimeter?

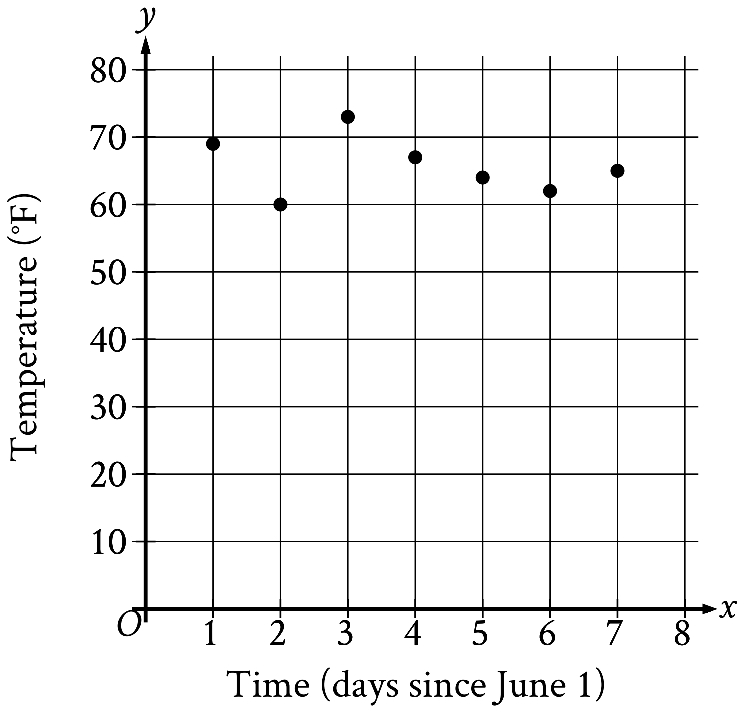
A. 3.6

B. 4.1

C. 4.6

D. 5.5

The scatterplot shows the temperature y, in °F, recorded by a meteorologist at various times x, in days since June 1.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending approximately horizontally.
- The data points have the following coordinates:
  - (1 comma 69)
  - (2 comma 60)
  - (3 comma 73)
  - (4 comma 67)
  - (5 comma 64)
  - (6 comma 62)
  - (7 comma 65)

During which of the following time periods did the greatest increase in recorded temperature take place?

A. From $x = 6$ to $x = 7$

B. From $x = 5$ to $x = 6$

C. From $x = 2$ to $x = 3$

D. From $x = 1$ to $x = 2$